

Project Design Phase

Proposed Solution Template

Date	04 September 2026
Team ID	
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	Commuters currently rely on manual, counter-based or physical ticketing systems for metro travel, which leads to long queues, delays, human errors in fare calculation, and inefficient use of station staff. There is no unified digital platform that allows passengers to book metro tickets instantly, calculate fares automatically, and receive a verifiable digital ticket without visiting a counter.
2	Idea / Solution Description	A "Metro Ticket Generating System" built entirely on the ServiceNow platform, using the Service Catalog, Flow Designer, Catalog Client Scripts, and Service Portal Widgets. Passengers select their source and destination stations and journey type (single/return) through a Service Catalog item ("Book A Metro Ticket"). The system auto-calculates the fare based on the selected route using client scripts, accepts digital payment mode selection, and instantly generates a scannable QR-coded digital ticket via a custom Service Portal widget - eliminating the need for physical tickets or counters.
3	Novelty / Uniqueness	Unlike traditional ticketing kiosks or third-party apps, this solution is built natively on a no-code/low-code enterprise workflow platform (ServiceNow) - combining Service Catalog automation, dynamic fare-calculation client scripts, conditional UI Policies, and real-time QR code generation within a single unified request-fulfillment workflow, without relying on external ticketing software.
4	Social Impact / Customer Satisfaction	Reduces passenger wait times at ticket counters, minimizes human error in fare collection, and provides a faster, contactless, and accessible ticket-booking experience. It also supports environmental sustainability by cutting down paper ticket usage and encourages digital adoption among commuters of all age groups.

S.No .	Parameter	Description
5	Business Model (Revenue Model)	Revenue is generated directly through digital ticket fare collection (per single/return journey, priced per route and passenger count) via integrated digital payment modes (UPI, Card, and other online modes). The platform can be extended to metro operators as a licensed digital ticketing add-on integrated with their existing ServiceNow ITSM/enterprise instance, reducing operational costs from manual ticketing infrastructure.
6	Scalability of the Solution	The solution can be scaled to support multiple metro lines, stations, and cities by extending the station/fare reference tables. Future scope includes SMS-based and Email-based ticketing, integration with real payment gateways (replacing the current dummy URL/QR logic), and a travel-analytics dashboard for operators to track passenger volume, peak hours, and route-wise revenue - as outlined in the project's "Scalability & Future Plan".